

GasLib-134

API & ASME B31.3 Technical Reference Document

Pipeline Network Design | Process Piping Standards | Compliance Guide

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Network	GasLib-134 Benchmark

1. GASLIB-134 NETWORK OVERVIEW

1.1 Introduction to GasLib-134

GasLib-134 is a widely referenced benchmark instance for natural gas transmission network optimization. The network contains 134 nodes (including supply sources, intermediate junctions, and consumer demand points) interconnected by pipeline segments, compressor stations, valves, and regulators. It is used extensively in academic and industrial research to validate steady-state and transient gas flow simulation algorithms.

The dataset was originally developed to facilitate benchmarking of mixed-integer nonlinear programming (MINLP) solvers and is distributed under the GasLib library. It represents a realistic medium-to-large scale gas network topology that exercises a broad range of engineering constraints relevant to both API and ASME standards.

1.2 Network Topology Summary

Component	Count	Description
Nodes (total)	134	Source, sink, and intermediate junctions
Source Nodes	6	Entry points with fixed supply pressure
Sink (Demand) Nodes	83	Consumer offtake points
Inner Nodes	45	Pipeline junctions / manifolds
Pipeline Segments	86	Pressure-driven flow conduits
Compressor Stations	22	Mechanical compression units
Control Valves	8	Regulating / pressure-reducing valves
Resistors	1	Fixed-resistance flow elements

Table 1 – GasLib-134 component inventory

1.3 Physical Parameters

Parameter	Value / Range	Unit
Gas composition (methane)	~95 %	mol %
Wobbe Index (approx.)	48.0 – 51.5	MJ/m³
Operating temperature	10 – 45	°C
Inlet pressure (max)	100	bar-g
Minimum delivery pressure	30	bar-g
Pipe diameters	100 – 1 200	mm (NPS 4 – 48)
Pipe lengths	0.5 – 120	km
Max flow rate	~15 000	MNm³/day

Table 2 – Representative physical parameters of GasLib-134

2. API STANDARDS APPLICATION

The American Petroleum Institute (API) publishes a suite of standards governing pipeline design, material selection, inspection, and operational integrity for gas transmission systems. The following sections identify the primary API documents applicable to a network of the GasLib-134 type.

2.1 API 5L – Line Pipe Specification

API 5L specifies requirements for the manufacture of two product specification levels (PSL 1 and PSL 2) of seamless and welded steel line pipe for use in pipeline transportation systems. For the high operating pressures present in GasLib-134 (up to 100 bar-g), PSL 2 pipe with enhanced mechanical testing and chemical composition limits is mandatory.

Grade	Min. Yield (MPa)	Min. Tensile (MPa)	Max Wall Thickness (mm)	Applicable NPS
X52	359	455	25.4	NPS 4–12
X60	414	517	31.8	NPS 6–24
X65	448	531	38.1	NPS 12–36
X70	483	565	44.5	NPS 18–48
X80	552	621	50.8	NPS 24–48

Table 3 – API 5L PSL 2 grade summary for GasLib-134 pipe selection

2.2 API 1111 – Offshore Pipelines (Reference)

Although GasLib-134 is modelled as an onshore network, API 1111 (Design, Construction, Operation, and Maintenance of Offshore Hydrocarbon Pipelines) provides valuable design factors for high-pressure gas pipelines. The design-factor concept ($F = 0.60 - 0.72$) from API 1111 is sometimes referenced in conjunction with ASME B31.3 for conservative wall-thickness calculations.

2.3 API 14E – Piping for Offshore Platforms

For compressor station piping within GasLib-134, API 14E governs the sizing of process piping to prevent erosion. The recommended erosional velocity (V_e) is calculated as:

$$V_e = C / \sqrt{\rho_m} \text{ where } C = 100 \text{ (continuous service) or } C = 125 \text{ (intermittent service), and } \rho_m \text{ is the gas-liquid mixture density (kg/m}^3\text{).}$$

2.4 API 6D – Pipeline Valves

The 8 control valves modelled in GasLib-134 must meet API 6D (Specification for Pipeline and Piping Valves). Key requirements applicable to this network include:

- Pressure class rating: ANSI Class 600 or 900 for pressures up to 100 bar-g
- Seat leakage class: Class VI per API 598 for control valves
- End-to-end dimensions: conforming to API 6D Table 1 for full-bore ball valves
- Fire-safe design: API 6FA or ISO 10497 certification required
- Material traceability: full material test reports (MTR) per API 5L / ASTM

2.5 API 520 / API 521 – Pressure Relief & Flare Systems

Each compressor station in GasLib-134 requires pressure-relief device (PRD) sizing per API 520 Part I (Sizing) and Part II (Installation), with blow-down and flare system design per API 521. Relief valve set pressure is typically 110 % of maximum allowable operating pressure (MAOP).

Station Type	MAOP (bar-g)	Relief Set Pressure (bar-g)	Min. Orifice Area (cm ²)
Compressor suction	50	55	3.2
Compressor discharge	100	110	8.6
Pressure regulating	60	66	4.5
Consumer delivery	30	33	2.1

Table 4 – Indicative PRD sizing parameters for GasLib-134 stations

3. ASME B31.3 – PROCESS PIPING

ASME B31.3 (Process Piping) governs the design, materials, fabrication, assembly, inspection, examination, and testing of process piping systems. Within GasLib-134, ASME B31.3 is the primary code for all above-ground station piping (compressor stations, metering skids, and valve assemblies). Long-distance buried transmission piping typically falls under ASME B31.8 (Gas Transmission), but B31.3 governs all station boundaries.

3.1 Scope and Fluid Service Classification

Natural gas at GasLib-134 operating conditions is classified as **Normal Fluid Service** under ASME B31.3 para. 300.2, because it does not meet the criteria for Category D (low hazard), Category M (highly toxic), or High-Pressure fluid service. This classification drives the required examination level and testing requirements.

Fluid Service	Pressure (bar-g)	Temperature (°C)	B31.3 Category	Exam Level
Fuel gas (low press.)	< 10	Ambient	Category D	Visual only
Instrument gas	< 15	Ambient	Category D	Visual only
Process gas (suction)	Up to 50	10 – 45	Normal	Random 10 %
Process gas (discharge)	Up to 100	10 – 60	Normal	Random 10 %
High-pressure seal gas	> 100	10 – 60	High Pressure	100 % RT/UT

Table 5 – Fluid service classification for GasLib-134 station piping

3.2 Wall Thickness Calculation (Para. 304.1.2)

The required minimum wall thickness t (mm) for straight pipe under internal pressure is given by:

$$t = (P \times D) / (2 \times (S \times E + P \times Y)) + c$$

Symbol	Description	Typical Value (GasLib-134)
P	Internal design pressure	Up to 100 bar-g (10 MPa)
D	Outside diameter of pipe	114.3 – 1 219 mm
S	Basic allowable stress (ASME B31.3 Table A-1)	138 – 241 MPa (X52–X65)
E	Longitudinal joint quality factor	1.0 (seamless), 0.85 (SAW)
Y	Temperature coefficient (para. 304.1.1)	0.4 ($T \leq 482$ °C)
c	Corrosion + mechanical allowance	1.5 – 3.0 mm

Table 6 – Wall thickness formula variables (ASME B31.3 para. 304.1.2)

3.3 Example Wall Thickness – NPS 24 Discharge Header

For the compressor discharge header (NPS 24, X65 seamless pipe, $P = 100$ bar-g, corrosion allowance $c = 3$ mm):

Step	Calculation	Result
Convert pressure	$100 \text{ bar-g} \times 0.1 = 10 \text{ MPa}$	$P = 10 \text{ MPa}$
Outside diameter	NPS 24 per ASME B36.10	$D = 609.6 \text{ mm}$

Step	Calculation	Result
Allowable stress	ASME B31.3 Table A-1, A358 Gr. X65	S = 241 MPa
Apply formula	$t = (10 \times 609.6) / (2 \times (241 \times 1.0 + 10 \times 0.4))$	t = 12.3 mm
Add corrosion	12.3 + 3.0	t_min = 15.3 mm
Select schedule	API 5L PSL 2, WT = 17.5 mm	Schedule 30 selected

Table 7 – Wall thickness worked example (GasLib-134 discharge header)

4. ASME B31.3 — MATERIALS, JOINTS & TESTING

4.1 Permissible Materials (Table A-1)

ASME B31.3 Appendix A lists allowable stresses for materials at temperature. For GasLib-134 operating temperatures (10–60 °C), the following materials are qualified:

ASME Material Spec.	Product Form	Grade / Type	Allowable Stress (MPa)	Remarks
ASTM A106	Seamless pipe	Gr. B	138	For NPS ≤ 20
ASTM A333	Seamless pipe	Gr. 6	138	Low-temp service
API 5L PSL2	Seamless/SAW	X52 – X80	172–241	Preferred for HP
ASTM A234	Fittings	WPB / WPC	138	Butt-weld fittings
ASTM A105	Flanges/fittings	—	138	Carbon steel flanges
ASTM A182	Flanges	F316L	115	Stainless, wet gas

Table 8 – Selected B31.3 allowable materials for GasLib-134

4.2 Welding and Examination Requirements

All welding must be performed in accordance with ASME Section IX (Welding and Brazing Qualifications). The examination extent depends on fluid service category:

Weld Type	Normal Service	High-Pressure Service	Category D
Butt welds (girth)	10 % RT or UT	100 % RT or UT	Visual
Socket welds	5 % MT/PT	20 % MT/PT	Visual
Branch connections	10 % MT/PT	100 % RT/UT	Visual
Fillet welds	5 % MT/PT	10 % MT/PT	Visual

Table 9 – NDE requirements per ASME B31.3 para. 341

4.3 Pressure Testing (Para. 345)

Upon completion of fabrication, all piping must be hydrostatically or pneumatically tested. For GasLib-134 station piping:

Test Type	Test Pressure	Hold Duration	Acceptance Criteria
Hydrostatic (preferred)	1.5 × MAOP × LS factor	Min. 10 minutes; typically 4 hours	No visible leakage or pressure drop
Pneumatic (approved alternative)	1.1 × MAOP	Min. 10 minutes	Soap-bubble or leak-detection fluid
Initial service test (Cat D only)	Operating pressure	While in service	No visible leakage

Table 10 – Pressure testing requirements per ASME B31.3 para. 345

4.4 Expansion and Flexibility (Para. 319)

For compressor station piping subject to thermal cycling, formal stress analysis per ASME B31.3 para. 319.4 is required when the simplified screening criterion is not satisfied. The displacement stress range S_E must not exceed

the allowable displacement stress range S_A :

$$S_E \leq S_A = f \times (1.25 S_c + 0.25 S_h)$$

where S_c and S_h are the basic allowable stresses at cold and hot conditions respectively, and f is the stress-range reduction factor ($f = 1.0$ for $N \leq 7\,000$ cycles).

5. COMPRESSOR STATION DESIGN (API + ASME B31.3)

GasLib-134 contains 22 compressor stations. The combined application of API standards (for machinery and ancillaries) and ASME B31.3 (for process piping within the station boundary) is critical to safe and efficient operation.

5.1 API 618 – Reciprocating Compressors

API 618 covers reciprocating compressors for petroleum, chemical, and gas industry services. Design approach selection (1, 2, or 3) depends on the pulsation and vibration analysis requirements:

- Design Approach 1 – simplified pulsation control via acoustic filters; suitable for low-speed, small-bore compressors
- Design Approach 2 – vendor-specified acoustic simulation; mandatory for compressors > 75 kW
- Design Approach 3 – full digital acoustic and mechanical simulation; required for compressors > 375 kW or where piping is sensitive to vibration

5.2 API 617 – Centrifugal Compressors

For high-flow compressor stations in GasLib-134, multi-stage centrifugal compressors per API 617 are the preferred technology. Key design requirements include rotor dynamic analysis (lateral and torsional), surge control system design (anti-surge valve per API 670 vibration monitoring), and mechanical running test at the manufacturer's facility.

5.3 Station Piping – ASME B31.3 Interface

Area	Governing Code	Design Basis
Buried transmission main	ASME B31.8	Location class, design factor
Station isolation (block valves)	API 6D / B31.8	MAOP × 1.25 test
Station inlet piping (scrubbers)	ASME B31.3	Normal fluid service
Compressor suction/discharge	ASME B31.3	10 % RT girth welds
Cooler and scrubber piping	ASME B31.3	Normal fluid service
Instrument and control piping	ASME B31.3	Category D service
Fuel gas supply to drivers	ASME B31.3	Category D service
Flare/blow-down header	API 521 / B31.3	Calc per API 521

Table 11 – Code demarcation for GasLib-134 compressor station piping

6. GAS FLOW EQUATIONS & HYDRAULIC CONSTRAINTS

6.1 Weymouth Equation (API TR 14E Reference)

The Weymouth equation is commonly used in GasLib-134 flow simulations for turbulent gas flow in high-Reynolds-number regimes:

$$Q = 3.7435 \times 10^{-3} \times T_b/P_b \times [(P_1^2 - P_2^2) \times D^{8/3} / (G \times T_f \times L \times Z)]^{0.5}$$

6.2 Panhandle A & B Equations

The Panhandle equations are applicable for partially turbulent flow typical of larger-diameter high-pressure transmission lines present in GasLib-134. Panhandle B (fully turbulent) is preferred for pipelines with Reynolds numbers $Re > 11 \times 10^6$.

6.3 Hydraulic Constraint Summary

Constraint	Symbol	Limit	Source
Max. inlet pressure	$P_{in,max}$	100 bar-g	ASME B31.8 / API 5L
Min. delivery pressure	$P_{del,min}$	30 bar-g	GasLib-134 spec
Compression ratio limit	r_c	≤ 2.0	API 617 / thermodynamic
Gas velocity (max)	V_{max}	20 m/s	Erosion limit (API 14E)
Min. flow (surge margin)	Q_{min}	$\geq 110\%$ surge	API 617 / control
Max. discharge temperature	T_{dis}	60 °C	Pipe coating limit
Line packing limit	$\Delta P/\Delta L$	0.05 bar/km	GasLib operational

Table 12 – Hydraulic constraints applicable to GasLib-134

7. INSPECTION, INTEGRITY MANAGEMENT & COMPLIANCE

7.1 API 570 – Piping Inspection Code

All station process piping in GasLib-134 must be managed under an inspection programme consistent with API 570 (Piping Inspection Code: Inspection, Repair, Alteration, and Rerating of In-Service Piping Systems). The code establishes:

- Inspection interval classification: Class 1 (high consequence) – max 5 years; Class 2 – max 10 years; Class 3 – max 10 years
- Thickness measurement locations (TML) to be identified and monitored for corrosion rate determination
- Remaining life calculation: $t_{\text{remaining}} = (t_{\text{actual}} - t_{\text{required}}) / C_r$
- Risk-based inspection (RBI) per API 580/581 as an alternative to prescriptive intervals

7.2 API 1160 – Managing System Integrity (Gas Pipelines)

For the buried pipeline segments in GasLib-134, API 1160 (Managing System Integrity for Hazardous Liquid Pipelines, adapted for gas networks in conjunction with 49 CFR Part 192) mandates:

- High-consequence area (HCA) identification and mapping
- Integrity assessment methods: ILI (in-line inspection), pressure testing, or direct assessment (ECDA/ICDA)
- Baseline assessment within 10 years of programme establishment
- Re-assessment intervals: 7 years maximum or per risk analysis

7.3 ASME B31.3 Re-rating and Repair

When operational changes in GasLib-134 require increasing MAOP or operating temperature beyond original design, the re-rating procedure per ASME B31.3 para. 345.9 must be followed, including:

- Review of all design calculations with the new operating conditions
- Confirmation that all installed components (valves, fittings, flanges) are rated for the new conditions
- Hydrostatic test at $1.5 \times$ new MAOP or NDE examination per para. 341
- Permanent update of piping records and P&IDs;

8. DESIGN COMPLIANCE CHECKLIST

The following checklist summarises key verification items for GasLib-134 pipeline and station design against API and ASME B31.3 requirements.

#	Compliance Item	Reference	Status
1	Line pipe material grade confirmed (API 5L PSL 2)	API 5L §7	■ Verified
2	Wall thickness $\geq$ calculated $t_{\min}$ + corrosion allowance	B31.3 §304	■ Verified
3	Fluid service category assigned to all piping circuits	B31.3 §300.2	■ Verified
4	NDE extent defined per service category	B31.3 §341	■ Verified
5	Pressure test procedure documented and approved	B31.3 §345	■ Verified
6	Pipe stress analysis completed (CAESAR II or equiv.)	B31.3 §319	■ Verified
7	PRD sizing completed per API 520 for each compressor	API 520 §3	■ Verified
8	Erosional velocity check completed (API 14E)	API 14E §2.5	■ Verified
9	Valve API 6D datasheet reviewed and approved	API 6D §5	■ Verified
10	Compressor pulsation study (API 618 DA selection)	API 618 §2.4	■ Verified
11	Weld procedures qualified per ASME Section IX	B31.3 §328	■ Verified
12	Flange ratings confirmed (ASME B16.5 / B16.47)	B31.3 §302.2.1	■ Verified
13	Inspection intervals established per API 570	API 570 §6	■ Verified
14	Integrity management programme per API 1160	API 1160 §5	■ Verified
15	As-built documentation and MTRs archived	B31.3 §346	■ Verified

Table 13 – API / ASME B31.3 compliance checklist for GasLib-134